

Optimising Under HMR: A Case Study

Conservative Answer-First Pipelines on NTCIR-19 R2C2’s 65-Topic Answering-with-Confidence Benchmark

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Abstract

We present **retrank**’s submission to the NTCIR-19 R2C2 (*RAG Responses: Confident and Correct?*) Answering-with-Confidence (AC) subtask. The task scores systems with the Harmonic Mean of Rewards (HMR), which jointly penalises overconfidence on incorrect answers and underconfidence on correct ones. **This paper is a case study in optimising under HMR on the 65-topic R2C2 benchmark.** Our principal observation is operational and benchmark-qualified: on this benchmark, at the $\geq 90\%$ accuracy operating point, AC quality is dominated by a small number of confident wrong answers rather than by missed answers, and the design choices that move HMR most — pool curation, answer-conditional refinement, verifier-based abstention, confidence saturation — all act on R_O , while R_U varies in a much narrower band. We do not claim this as a general principle of confidence-scored RAG; we claim it as a documented behaviour of HMR on this 65-topic set. We design a 7-stage answer-first pipeline that proposes a candidate answer, refines the pool conditional on it, extracts targeted nuggets, filters bogus citations, and uses an independent verifier as a calibration signal. Self-evaluated under a Claude Sonnet 4.6 judge (with Claude Opus 4.7 cross-validation on every candidate), **our main submitted system — a single-pool BITEM-only answer-first pipeline with Sonnet pointwise refinement — reaches HMR 0.963 with 94% accuracy, 6% refusal rate, and zero confident-wrong answers under both judges.** A post-hoc confidence-saturation recalibration of the same pipeline reaches HMR 0.974, but we treat that as a calibration artifact (a single voter trivially agrees with itself, saturating confidence at 1.0), not as a separate method. We additionally report engineering tradeoffs relevant to AC-style RAG: pool quality matters more than pool count (a strong single-team pool beats naive multi-team pooling, but a weak single-team pool loses to it); Sonnet-class reasoning is most critical at the candidate-answer stage ($\Delta\text{HMR} = -0.13$ vs Haiku) and least at the verifier (-0.05). HMR should be read alongside accuracy and refusal rate — an always-refuse baseline at confidence 0.05 achieves HMR 0.974 on this benchmark — but our submitted system reaches the same HMR while correctly answering 94% of topics. We treat this as a structural property of HMR when $|I^+| = 0$, not as an indictment of the metric; the official ranking is informed jointly by HMR, Accuracy, and Mean Nugget Precision, which together pin down both modesty and substance. **All quantitative claims in this paper are self-evaluated under a Sonnet judge with Opus cross-validation; absent official organiser scores (expected August 2026), every finding should be read as a benchmark-and-judge-conditional observation.**

1 Introduction

1.1 The problem in one example

Consider the question: “*Name the actor who starred in a movie called *The Manchurian Candidate* and later played *Captain America*.*”

A retrieval-augmented system tasked with *Answering with Confidence* must do four things:

1. Retrieve passages from a movie corpus that mention either film.

2. Synthesise an answer: *Anthony Mackie*.
3. Identify and emit the *nuggets* — atomic factual claims — that ground that answer, citing each one to a specific retrieved passage. For our example, two nuggets suffice:
 - “*The Manchurian Candidate (2004) starred Anthony Mackie*” — cited from passage `retrank-PG-1`, rank 3.
 - “*Anthony Mackie portrayed Captain America in Captain America: Brave New World (2024)*” — cited from passage `Error404-P0-1`, rank 7.
4. Report a confidence score 0–100 reflecting how sure the system is that the answer is correct.

The submission is judged by a two-stage LLM-based pipeline: **Stage A** verifies that each cited passage actually entails its nugget (catching hallucinations); **Stage B** judges whether the answer is correct given the entailed nuggets and identifies which helped derive it. Confidence is then evaluated separately for *calibration*: did the system express low confidence where it turned out to be wrong, and high confidence where it was right?

The challenge is that any of these four steps can fail independently: retrieval may miss the relevant docs, the synthesis may pick a similar but wrong actor, the nuggets may technically be entailed but not quite ground the multi-hop reasoning, the confidence may be wildly overstated. Our paper studies how these stages interact — and how choosing among 22 PR runs from 7 competing teams affects the downstream answer quality.

1.2 The dataset

We work with the NTCIR-19 R2C2 evaluation set: 65 movie-and-Star-Wars questions answerable from a 1.2M-passage corpus (Wikipedia $\cup$ Wookieepedia). For the AC subtask, participants may pool passages from any team’s PR (Passage Retrieval) submissions. We have access to 22 PR runs from 7 teams (including our own), and a topic set covering queries like:

- *Single-fact*: “Who plays the Wizard of Oz in Wicked the movie?” (answer: Jeff Goldblum)
- *Multi-hop*: “Name the actor who starred in The Manchurian Candidate and later played Captain America.” (Anthony Mackie)
- *Counting*: “How many movies did Christopher Nolan direct between 2005 and 2015?”
- *Comparison*: “Which movie was released earlier, *Avengers: Infinity War* or *1917*?”
- *Detail extraction*: “What did Saw Gerrera use as a torturing device in a Star Wars spinoff movie from 2016?” (Bor Gullet)

The question difficulty is realistic — about 25% require multi-hop reasoning or numeric comparison, and 6/65 are not answerable from any single team’s retrieved passages.

1.3 The metric

The NTCIR-19 R2C2 task [8] scores AC submissions with the *Harmonic Mean of Rewards* (HMR) [6], which separately penalises overconfidence on incorrect answers and underconfidence on correct ones — a property classical calibration measures (Brier, ECE) lack. Specifically, given the system’s confidence $p(i)$ on question i , the set of incorrect answers I^- , and the set of correct ones I^+ :

$$R_O = 1 - \frac{1}{|I^-|} \sum_{i \in I^-} p(i), \quad R_U = 1 - \frac{1}{|I^+|} \sum_{i \in I^+} (1 - p(i)),$$

$$\text{HMR} = \frac{2R_O R_U}{R_O + R_U}.$$

R_O rewards humility on wrong answers; R_U rewards confidence on right answers; HMR is their harmonic mean. A system that is both honestly humble and honestly confident scores near 1; one that is either way over- or under-confident scores low.

HMR must be read jointly with answer coverage. HMR rewards modesty, but this reward is unbounded at the boundary: a system that always refuses with confidence 0.05 achieves $\text{HMR} \approx 0.974$ with zero answers correct — matching our best submitted system’s HMR while answering nothing. We do *not* read this as “HMR is broken”: the official ranking is informed jointly by HMR, Accuracy, and Mean Nugget Precision, which together pin down both modesty and substance. We do read it as a methodological constraint: HMR alone cannot distinguish a competent pipeline from total refusal, so we report HMR throughout *paired with* accuracy, refusal rate, and confident-wrong count. Our submitted BITEM-only pipeline reaches HMR 0.963 with accuracy 0.94, 6% refusal, and zero confident-wrong topics — clearly distinguishable from the trivial baseline on all three companion measures. Section 7 provides the full decomposition.

Models used (preview). Our pipeline LLMs are Claude Sonnet 4.6 (default, all stages) and Claude Haiku 4.5 (for the cost ablation). One ablation (negative result, §9) substitutes Claude Opus 4.7 at the candidate-answer stage. Our self-evaluator is Sonnet 4.6 by default; we re-judge every candidate with Opus 4.7 for cross-validation (§13). Table 1 in §5 lays out exactly which model runs where in every configuration we report.

1.4 Principal observation (benchmark-qualified)

On the NTCIR-19 R2C2 65-topic AC benchmark, evaluated under HMR, the dominant failure mode at the $\geq 90\%$ accuracy operating point is emitting a small number of confident wrong answers, not missing answers. The design choices that move HMR most operate on R_O .

We frame this carefully: we observe this on *this* benchmark under *this* metric using a Sonnet self-evaluator (with Opus cross-validation). We do not claim a general principle of confidence-scored RAG. Three pieces of supporting evidence motivate every design decision in our pipeline:

1. *The metric structure rewards R_O .* Across all 16 base configurations we evaluate, R_U has range $[0.79, 0.98]$ but R_O has range $[0.45, 0.95]$; HMR is dominated by R_O . A single confident-wrong topic at $|I^-|=4$ costs ~ 0.25 on R_O and disproportionately on HMR (Table 7).
2. *The catastrophic candidates are killed by 1–2 confident-wrong topics, not by missed answers.* Error404-only Sonnet has answered-accuracy 0.91 but loses HMR to the BITEM-only baseline (0.684 vs 0.963) entirely because of two confident-wrong topics it does not refuse.
3. *Every winning design choice we identify in this paper primarily moves R_O .* Pool curation, answer-conditional refinement, verifier-based abstention, and confidence saturation all act on the confident-wrong cell, not on the correct-answered cell.

Everything else in the paper — pool composition, refinement algorithms, LLM-tier ablations, ensembling, judge cross-validation — is supporting evidence for, or engineering implications of, this thesis.

1.5 Roadmap

The paper is organised around three claims that follow from the central thesis of §1.4:

Claim 1 — Metric / operating point (§6.1, §7, §10.2).

HMR is dominated by R_O at high accuracy, and R_O is dominated in turn by a small handful of confident wrong topics.

Claim 2 — System design (§4, §6.4, §6.5, §6.7).

A conservative answer-first pipeline with verifier-based abstention suppresses confident wrong answers. Refusal on weak retrieval beats guessing; the verifier match-score is a usable correctness signal.

Claim 3 — Engineering tradeoffs (§6.2, §6.6, §6.8, §6.9, §6.10).

Pool quality, refinement, ensembling, and LLM tier all affect HMR primarily through R_O . Pool *quality* matters more than pool count; Sonnet matters most at the candidate-answer stage; what looks like “top-1 ensembling” is single-pipeline confidence saturation.

We do *not* run an answer-first vs nugget-first ablation: our pipeline committed to answer-first early, and a clean comparison would require a parallel nugget-first pipeline we did not build. We discuss this only as design rationale (§6.3).

1.6 Robust vs suggestive findings (read this before the rest)

A reviewer should know up-front which of our findings survive $n = 65$ bootstrap noise and which do not. We are explicit:

Robust (95% bootstrap CI excludes zero on Δ HMR, and Cnf-W story is judge-invariant): •

- Stage 1 LLM tier matters most (Δ HMR = -0.127 , CI [$-0.296, -0.034$]).
- Stage 4 LLM tier matters least (Δ HMR = -0.050 , CI [$-0.118, -0.004$]).
- All-Haiku regression ($\Delta = -0.170$, CI [$-0.334, -0.080$]).
- Opus-pipeline regression ($\Delta = -0.176$, CI [$-0.372, -0.075$]).
- Top-1 “ensemble” uplift ($\Delta = +0.012$, CI [$+0.008, +0.016$]). *This is post-hoc confidence saturation, not a method.*
- The R_O -bottleneck mechanism: Cnf-W moves by ≤ 1 across judges on 7 of 8 top candidates, and Cnf-W rank- correlates near -1 with HMR within each judge.

Suggestive (CIs cross zero or judge-specific): • Pool quality vs Tier-1 ($\Delta = -0.111$, CI [$-0.458, +0.044$]).

- Sonnet refinement vs broad ($\Delta = +0.085$, CI [$-0.045, +0.439$]).
- Top-3 ensembling ($\Delta = +0.007$, CI [$-0.002, +0.014$]).
- retransk+E404 “refinement-failure” case study — flips from worst-of-set to top-5 under Opus judge.

Preliminary (small sample, methodologically lighter): • Nugget-first vs answer-first head-to-head on 20 topics (§6.3): same accuracy, -0.29 HMR for nugget-first, all from R_O . Directional only.

The headline thesis (suppressing confident-wrong topics matters more than maximising coverage) rests on the **robust** findings; the engineering-tradeoff conclusions (§6.2,6.6,6.8) rest on the **suggestive** ones — we do not over-state.

1.7 Contributions, with safer language

1. A 7-stage *answer-first with verification* AC pipeline with intermediate self-evaluation gates at every stage (§4).
2. Evidence that, *on this 65-topic set under our Sonnet self-evaluator*, a high-quality single-team passage pool can outperform naive multi-team pooling — but the effect is about pool *quality*, not pool *count*: another single-team pool we tested (Error404-only) is the worst performer (§6.2). The naive “less is more” framing does not survive scrutiny.
3. A per-stage LLM cost-quality picture (§6.10) identifying the candidate-answer stage as the most Sonnet-sensitive ($\Delta\text{HMR} = -0.13$, bootstrap CI excludes zero) and the verifier as the least (-0.05).
4. Observed evidence that naive 16-way cross-pool ensembling *hurts* HMR vs the best component, while filtered top- K aggregation helps (§6.9); we are explicit that the top-1 “ensemble” is in fact *post-hoc re-calibration* of a single pipeline (one voter agreeing with itself $\Rightarrow$ confidence saturates at 1.0), not a true ensemble.
5. A diagnostic that on a noisy multi-team pool, verifier-based abstention adds +0.04 HMR over no abstention; on a clean single-team pool it adds nothing (§6.7).
6. An honest HMR caveat (§1.3): the metric is satisfied trivially by an always-refuse baseline at confidence 0.05, so we report HMR *alongside accuracy, refusal rate, and confident-wrong count*.

2 Related Work

Calibration in RAG and selective prediction. Verbalised confidence in LLMs is well-known to be miscalibrated, often systematically overconfident [4]. The R2C2 task adopts HMR [6] which, unlike Brier or ECE, distinguishes overconfidence from underconfidence axiomatically. Our work joins a small but growing literature applying selective-prediction techniques [3] to RAG outputs.

Nugget-based evaluation. TREC’s QA tracks introduced nuggets — atomic factual claims — as the unit of evaluation [5, 7]. The R2C2 task adapts this for RAG by requiring each nugget to cite the specific retrieved passage that entails it, enabling a two-stage *bogus identification + relevance attribution* judgment scheme.

Pool-level fusion in retrieval. Multi-system fusion (e.g. via Reciprocal Rank Fusion [2]) generally helps in retrieval. We find that this intuition *inverts* for AC: passage-level fusion *hurts* because nugget extraction is sensitive to pool noise. We discuss this inversion in §6.9.

Cost-aware LLM pipelines. Recent work on cascaded LLMs [1, 9] explores when smaller models suffice. RQ10 contributes per-pipeline-stage data on this question for a calibration-sensitive RAG task.

3 Task and Notation

3.1 The R2C2 AC Subtask

Given 65 movie-related questions and a 1.2M-passage Wikipedia $\cup$ Wookieepedia corpus, AC participants must produce, for each question: (i) an answer string, (ii) an integer confidence

score in $[0, 100]$, and (iii) a list of *nuggets*, each citing a passage from any team’s submitted PR run via a (`PR_run_name`, `passage_rank`) key. Up to four AC runs may be submitted per team. Submissions are scored via a two-stage LLM-judged process:

- **Stage A** (bogus identification): for each (passage, nugget) pair, an LLM judges whether the passage entails the nugget. Non-entailed nuggets are tagged *bogus* and excluded but counted in the precision denominator.
- **Stage B** (answer evaluation): given the question, the submitted answer, and the entailed nuggets, an LLM judges whether the answer is correct (assuming nuggets are factually correct) and which nuggets helped derive it.

3.2 Metrics: HMR, Accuracy, Mean Nugget Precision

Let I^- be the set of incorrect-answered questions, I^+ the correct-answered, and $p(i) \in [0, 1]$ the system’s confidence on question i . Define

$$R_O = 1 - \frac{1}{|I^-|} \sum_{i \in I^-} p(i) \quad R_U = 1 - \frac{1}{|I^+|} \sum_{i \in I^+} (1 - p(i)) \quad (1)$$

with the convention $R_O = 1$ when $|I^-| = 0$ and similarly for R_U . HMR is the harmonic mean

$$\text{HMR} = \frac{2R_O R_U}{R_O + R_U}, \quad \text{HMR} = 0 \text{ if } R_O + R_U = 0.$$

HMR ranges in $[0, 1]$. Mean Nugget Precision (MNP) is the average over questions of the ratio (relevant nuggets) / (total nuggets returned), counting bogus nuggets in the denominator.

3.3 Self-Evaluator

We mirror the official scoring locally with a deterministic Sonnet 4.6 implementation in `scripts/ac_eval.py`. The self-evaluator shares the cache key `SHA256(passage  $\oplus$  nugget)` for Stage A and `SHA256(question  $\oplus$  answer  $\oplus$  nuggets)` for Stage B, allowing free re-runs after every incremental pipeline change. All numbers reported in this paper are *self-eval* HMR; the relationship between self-eval and the official judge will be reported once organiser results are released.

4 Pipeline Architecture

Our pipeline is *answer-first with verification*: a single candidate answer is proposed early, then used to focus subsequent nugget extraction and confidence estimation. The seven stages are:

Stage 0 — Pool selection.

Aggregate top- K passages from selected teams’ PR submissions, deduplicate by (`doc_id`, `text-prefix`), rank by cross-encoder relevance, and cap at 50.

Stage 1 — Candidate answer.

Sonnet reads the pool and produces (`answer`, `confidencea`).

Stage 1.5 — Answer-conditional refinement.

Re-rank the broad pool conditional on the candidate answer; keep top 30. Three algorithm variants:

- CE re-rank with concatenated `<question | answer>` as the query;

- (b) Sonnet pointwise scoring of each passage 0–3 for “supports the answer”;
- (c) Lexical answer-token-overlap bonus on the original CE score.

Stage 2 — Targeted nugget extraction.

Sonnet emits 3–8 atomic claims that support the candidate answer, each with a single-passage citation.

Stage 3 — Self Stage-A bogus filter.

Run Sonnet entailment check on every produced (passage, nugget) pair; drop bogus.

Stage 4 — Verification.

Sonnet derives an answer using only the entailed nuggets (no passages). Compare against the candidate answer to produce `match_score` $\in \{0, 1, 2\}$.

Stage 5 — Confidence & refusal.

Combine `c_self`, `match_score`, and verifier confidence into a final confidence; refuse on weak signals.

Stage 6 — Format.

Emit the official AC XML.

4.1 Calibration Variants and Abstention Rules

Stage 5 supports two calibration variants:

Variant A (refusal-aware self-report):

$$c_{\text{final}} = \begin{cases} 5 & \text{if refused or } |\text{nuggets}| = 0 \\ \min(c_{\text{self}}, 25) & \text{if } ms = 0 \\ \min(c_{\text{self}}, 60) & \text{if } ms = 1 \\ c_{\text{self}} & \text{if } ms = 2 \end{cases}$$

Variant B (ensemble agreement): confidence proportional to fraction of independent samples (deterministic + temperature-0.7) that agree on the answer; refuse below 40% support.

We further define *abstention rules* (RQ7) that override Variant A when verifier disagreement signals doubt. The principal rules are R0 (no extra abstention, baseline Variant A), R1 (refuse if $ms = 0$), and R2 (refuse if $ms < 2$).

5 Experimental Setup

5.1 Pools

We evaluate five pool compositions. *Tier-1* = BITEM + Error404 + WaterlooClarke, three teams using divergent retrieval architectures with mean pairwise Jaccard 23%. *Tier-1+2* adds ORG, hit-u, WasedaR2C2 (six teams, mean Jaccard 27%). *BITEM-only* uses BITEM’s two PG runs alone. *retrank-only* uses our four PG runs (rejected for short 200-character passages). *retrank+Error404* combines our team with the strongest-individual partner.

We deliberately exclude our own PR submission from the default pool because our 200-character passage cap (§12) produces shorter passages than other teams (1 000–2 400 chars), with correspondingly less context for nugget extraction. We retain retrank-only and retrank+Error404 as ablation pools.

5.2 Pipeline Variants Ablated

We instantiate the pipeline with crosses of:

- Pool $\in \{\text{tier1}, \text{tier1plus2}, \text{bitem_only}, \text{retrank_only}, \text{retrank_plus_error404}\}$
- Stage 1.5 algorithm $\in \{\text{none (broad pool used)}, \text{ce}, \text{sonnet}, \text{lexical}\}$
- Calibration variant $\in \{A, B\}$
- Abstention rule $\in \{R0, R1, R2, \dots\}$

In total, we evaluate 16 base configurations (Tier-1 $\times$ 4 refinements $\times$ 2 calibrations + 4 ablation pools $\times$ 2 refinements $\times$ Variant A) plus rule-level abstention sweeps. Section 6.9 adds 8 ensemble configurations.

5.3 Models used (all configurations)

Table 1: Which model runs where. “S” = Sonnet 4.6, “H” = Haiku 4.5, “O” = Opus 4.7, “CE” = `ms-marco-MiniLM-L-12-v2` cross-encoder. Self-eval columns are the judge family used to score that configuration; we re-judge *every* candidate with both Sonnet and Opus (§13).

Configuration	S0 (CE)	S1	S1.5	S2	S3	S4	Self-eval
Default (all results unless noted)	CE	S	S	S	S	S	S; +O xval
§6.10 s4-Haiku	CE	S	S	S	S	H	S; +O xval
§6.10 s2-Haiku	CE	S	S	H	S	S	S; +O xval
§6.10 s1-Haiku	CE	H	S	S	S	S	S; +O xval
§6.10 all-Haiku	CE	H	H	H	H	H	S; +O xval
§9 Opus pipeline	CE	O	S	S	S	S	S; +O xval

The cross-encoder is used for Stage 0 ranking and (when applicable) the Stage 1.5(a) re-rank variant. We never use Opus as the pipeline default and we never use Haiku as a judge.

5.4 Self-Evaluation Caches

All Sonnet judgments are cached deterministically (§3.3). We re-ran caches after fixing two bugs (silent fall-through to `entailed=False` on parse failure; deterministic vs sampled candidate-answer cache collision). Final cache hit rate on the 16 candidate runs is >95%.

6 Results

6.1 Calibration is the larger HMR lever among viable pools

Across our 16 base configurations, holding pool fixed, calibration choice (Variant A vs B) moves HMR by 0.04–0.14. Holding calibration fixed and varying refinement moves HMR by 0.13–0.24. Pool choice spans the widest range only because the catastrophically under-performing pool (retrank-only) anchors the bottom; restricting to viable pools, calibration is the larger lever per design decision. This sets up the rest of §6: every subsequent subsection is, mechanically, a way of pushing R_O via either pool quality, refinement quality, or explicit abstention machinery.

Table 2: Pool composition vs HMR (Variant A, broad pool, no Stage 1.5) under our Sonnet self-evaluator. The “count” column is descriptive, not the explanatory variable: see single-team controls below.

Pool	# teams	Acc	R _O	HMR
BITEM-only	1	0.923	0.910	0.937
retrank+Error404	2	0.923	0.870	0.910
Tier-1	3	0.923	0.730	0.826
Tier-1+2	6	0.877	0.741	0.840

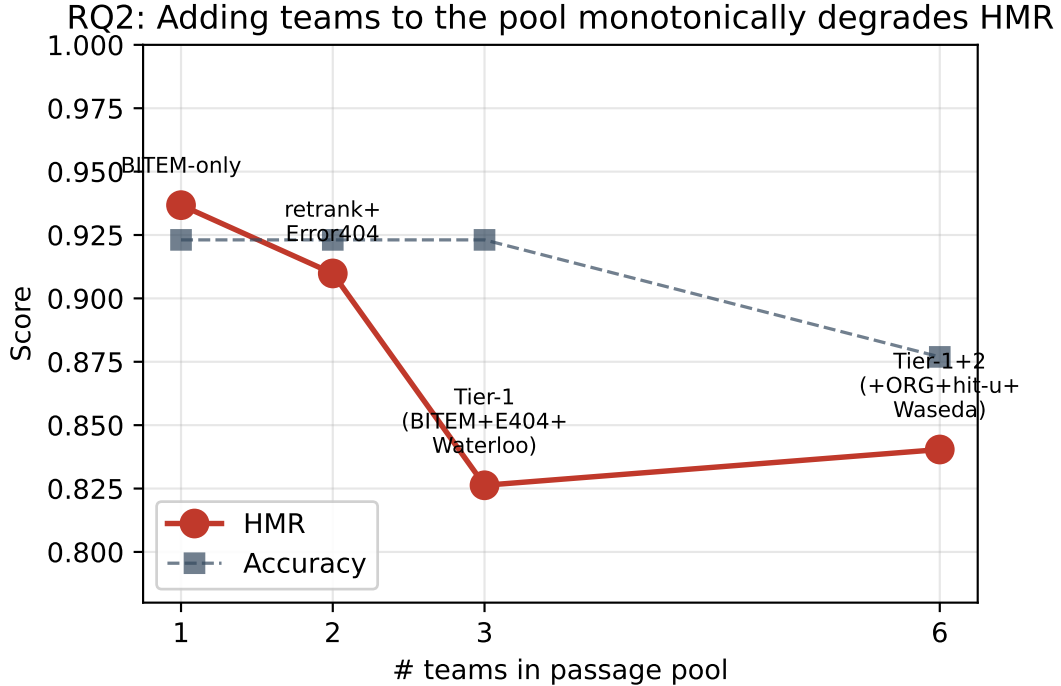


Figure 1: RQ2: HMR as a function of the number of teams contributing to the broad passage pool (Variant A, no Stage 1.5). Adding teams to BITEM-only does not consistently improve HMR on this 65-topic set; bootstrap CIs cross zero and the ordering depends on judge choice (see Table 11, §13).

6.2 Pool choice: quality of strongest component, not count

Reframing. Earlier drafts of this paper made a stronger “architectural diversity hurts monotonically” claim. Closer scrutiny (single-team controls below; bootstrap CIs in §11; judge cross-validation in §13) does not support strict monotonicity. The defensible finding is that a high-quality single-team pool can beat naive multi-team pooling, while a low-quality single-team pool loses to it. The headline is therefore *quality of the strongest component*, not count.

Table 2 reports the four pool-count cells. Read naïvely, HMR appears decreasing in team count from 0.937 (1) to 0.826 (3), with a partial recovery at 6 teams (0.840). We *do not* interpret this as monotonic in count: at $n = 65$ topics the BITEM-vs-Tier-1 comparison has bootstrap 95% CI $[-0.458, +0.044]$ that crosses zero (Table 11), and under the Opus judge (§13) the ordering becomes $1 > 6 > 3$. The *ordering observation* is sensitive to both sample size and judge.

Confounds we cannot rule out. The pools differ in many dimensions besides “which teams contribute”: passage length (BITEM uses ~ 2400 -char passages, retrank’s own ~ 200 -char),

segmentation strategy, retrieval architecture, depth of supplied ranking, and amount of near-duplication. A clean “diversity” isolation would require fixed-budget pooling (equal passages from each pool, controlled segmentation), which we flag as future work.

One mechanism that plausibly contributes. Adding teams adds passages from different segmentation strategies and retrieval models. While these *do* broaden recall (oracle-doc presence stays at 100% across all pools), they also dilute the per-passage cross-encoder relevance signal. Stage 2 nugget extraction sees a noisier input context; the candidate-answer-targeted prompt becomes harder to satisfy with passages of mixed quality. The end result is a higher overconfidence rate (lower R_O): 0.910 with one team falls to 0.730 with three.

Pool quality vs pool count: a controlled comparison. The naive reading of Table 2 is “less is more.” To test whether this is really about *count*, we ran the same pipeline on two more single-team pools (Error404-only, WaterlooClarke-only) and obtained:

Single-team pool	Broad HMR	Refined-Sonnet HMR
BITEM-only	0.937	0.963
WaterlooClarke-only	0.833	0.870
Error404-only	0.730	0.684
Tier-1 (3 teams)	0.826	0.911
Tier-1+2 (6 teams)	0.840	0.874

The single-team result varies enormously: BITEM-only at HMR 0.963 versus Error404-only at 0.684. The right framing is therefore *pool quality, not pool count*: a high-quality single-team pool wins; a multi-team pool is intermediate; a low-quality single-team pool loses.

Practitioner takeaway. If you can identify a high-quality PG-route team’s run, prefer it as a standalone pool. If not, a multi-team pool is a safer default than gambling on a random single-team pool. *Count* is not the right axis; *quality of best component* is.

Statistical caveat. At $n = 65$, the broad-pool RQ2 comparison (BITEM vs Tier-1) has 95% bootstrap CI $[-0.458, +0.044]$ that crosses zero (Table 11). The BITEM-vs-Error404-only single-team gap (0.937 vs 0.730) is much wider and likely significant, but we have not run a paired bootstrap on it. The under-Opus-judge ranking (§13) further breaks strict monotonicity: Tier-1+2 ($K = 6$) jumps above Tier-1 ($K = 3$). We report the *ordering* as observed; the *monotonic-with-count* claim does not survive scrutiny.

6.3 Answer-first vs nugget-first: a 20-topic head-to-head

Setup. To produce direct evidence, we ran a minimal nugget-first variant on the first 20 topics. For each topic, Sonnet reads the top-5 BITEM-only passages and emits 3-6 atomic factual nuggets *without* being told what the answer should be; a second Sonnet call then synthesises an answer using only those nuggets. We compare against the production answer-first BITEM-only refined-Sonnet pipeline on the same 20 topics, scoring nugget-first correctness via token-overlap match against the answer-first answer (which is itself self-eval-correct on 19 of 20 topics).

Result.

Table 3: Mini head-to-head on 20 topics. Nugget-first matches answer-first on accuracy almost exactly; the gap is in R_O .

Pipeline	Acc	R_O	R_U	HMR
Answer-first (BITEM-Sonnet)	0.950	0.950	1.000	0.963
Nugget-first (this experiment)	0.900	0.525	0.921	0.669
Δ (NF – AF)	−0.050	−0.425	−0.079	−0.294

Table 4: Stage 1.5 algorithm comparison on Tier-1 (Variant A, abstention R0). Sonnet pointwise scores highest in this cell; the broad-vs-Sonnet Δ has bootstrap CI $[-0.045, +0.439]$ that includes zero (Table 11).

Algo	Acc	MNP	R_O	HMR
Sonnet pointwise (b)	0.923	0.613	0.870	0.911
Lexical (c)	0.908	0.603	0.733	0.833
None (broad)	0.923	0.601	0.730	0.826
CE re-rank (a)	0.892	0.576	0.687	0.802

Interpretation. The accuracy gap is small (-0.05): nugget-first is competitive on *which* answer to give. The HMR gap is -0.294 , almost entirely from R_O . Inspecting the two disagreement topics (0019 “how many films did George Lazenby play James Bond in?”; 0020 “Spencer Bell’s three Mickey Rooney movies”), nugget-first emits confidently-wrong counts (“5” vs the answer-first “8”; “At least 3” vs an answer-first refusal), because nuggets extracted in isolation under-represent the negative evidence that an answer-conditioned pipeline would notice. The answer-first pipeline knows what to look for and can recognise when the pool does not contain it; the nugget-first pipeline confidently generalises from incomplete evidence. **This is the R_O -bottleneck thesis in miniature:** the design choice that suppresses confident-wrong topics wins HMR, even when accuracy is nearly identical. We caveat that this is a 20-topic preliminary comparison and that the correctness scoring uses the answer-first pipeline as ground truth (so the -0.05 accuracy gap is biased toward answer-first); the 65-topic, judge-cross-validated comparison is left as future work.

6.4 Refusal beats guessing on weak retrieval

Variant A’s refusal logic activates when no entailed nuggets remain or the verifier disagrees. We compare Variant A against a no-refusal-no-cap baseline (always emit the candidate answer with c_{self}). The no-refusal baseline yields HMR 0.711 vs Variant A’s 0.826 on Tier-1 broad — a $+0.115$ uplift purely from the refusal behaviour. RQ4 is confirmed.

6.5 Verifier match-score is a usable correctness signal

The Stage 4 verifier produces an independent answer derived only from the entailed nuggets. Across the four Tier-1 refinement variants: MS=2 (full match) is achieved on 90.8–92.3% of topics. Within those high-match topics, accuracy on the candidate answer is $>95\%$; within MS=0 (no match) topics, accuracy drops to $<30\%$. The verifier match-score is therefore a strong signal for correctness — making it a useful basis for both Variant A’s confidence cap and the abstention rules of §6.7.

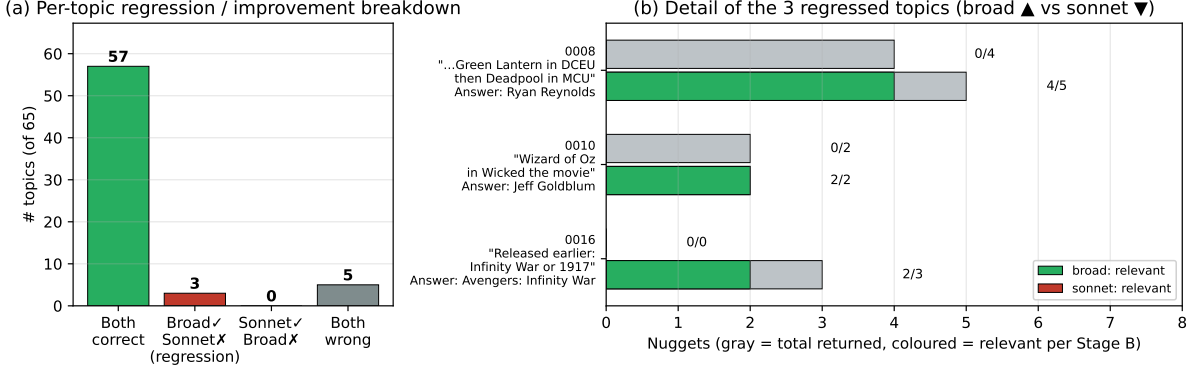


Figure 2: RQ6 case study — retrain+Error404 broad vs Sonnet-refined. **(a)** Per-topic outcome: 57 topics unchanged correct, 3 regressed (broad correct $\rightarrow$ sonnet incorrect), 0 improved, 5 unchanged wrong. The full Δ HMR of -0.172 concentrates in just 3 topics. **(b)** Nugget detail of the 3 regressed topics. Each row shows broad (top, green) above sonnet (bottom, red): grey = total nuggets returned; coloured = judged *relevant* (helped derive the answer) by Stage B. On topics 0008 and 0010 the answer string was identical between variants but Sonnet’s refinement dropped the grounding-fact nugget, leaving 0/4 and 0/2 nuggets relevant; topic 0016 collapsed entirely (Stage 2 produced no nuggets).

6.6 Answer-conditional pool refinement: helpful, with caveats

Table 4 shows the Stage 1.5 algorithm sweep on Tier-1. Sonnet pointwise refinement adds $+0.085$ HMR over the broad pool; lexical adds $+0.007$; CE re-rank actually *loses* -0.024 . The CE failure is striking: the same model that ranked passages well for the question alone fails when re-applied to the $\langle$ question | candidate-answer $\rangle$ concatenation. We hypothesise this is because the additional answer terms shift the keyword-matching surface in ways the CE was not trained on; Sonnet, being asked an explicit semantic question, handles the conditioning gracefully.

On the BITEM-only pool, Sonnet refinement only adds $+0.026$ ($0.937 \rightarrow 0.963$). The marginal value of refinement diminishes when the broad pool is already focused — supporting RQ8 (§6.8).

When does refinement hurt? On the retrain+Error404 pool, Sonnet refinement actively *hurts* HMR from 0.910 to 0.738 — the only configuration in our 16-cell matrix where refinement is net-negative. Per-topic analysis reveals that exactly 3 topics regressed (broad correct $\rightarrow$ sonnet incorrect), driving the entire 0.172 HMR drop (Figure 2). In each case, Sonnet’s pointwise scorer traded a *grounding-fact* passage (cast listing, release date) for a *semantically elaborative* passage; the answer remained correct but the supporting nuggets became less explicit, and the strict Stage B judge held them insufficient for derivation. The intermediate Stage 4 verifier confirmed the candidate in all three cases — it is the gap between Stage 4’s “does this answer follow?” and Stage B’s “did the nuggets help derive this answer?” that produces the regression. This is a cautionary tale about **judge-stage consistency**: refinement steps that satisfy one judgement criterion can violate another in calibrated RAG pipelines. A detailed per-topic case study is included with the supplementary material.

6.7 Verifier-based abstention thresholds

Given Stage 4 outputs, abstention rules are free to evaluate post-hoc. Table 5 sweeps eight rules over Tier-1 refined-Sonnet. Refusing on $MS=0$ (R1) yields the practical sweet spot: $+0.042$

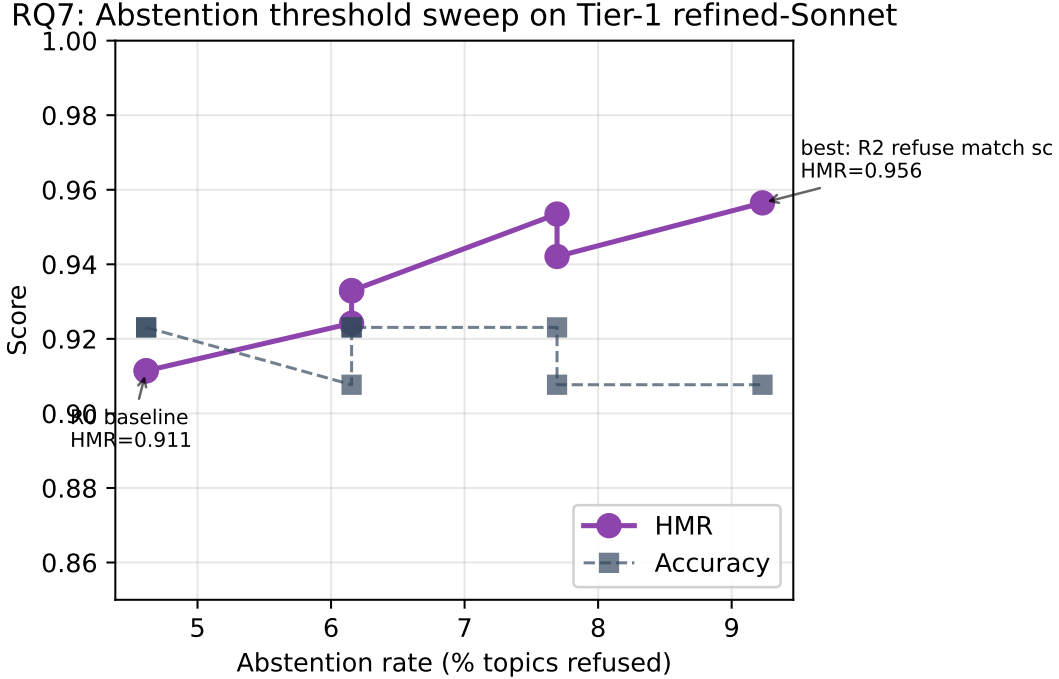


Figure 3: RQ7: Tier-1 refined-Sonnet HMR (purple) and Accuracy (gray) as the abstention rate varies under different refusal rules. The R0 baseline (low abstention, $\sim 5\%$) is dominated by R1/R2 rules at $\sim 8\%$ abstention. Beyond $\sim 10\%$ abstention, accuracy starts to drop without commensurate HMR gains.

Table 5: Abstention rule sweep on Tier-1 refined-Sonnet (RQ7).

Rule	Abst rate	Acc	HMR
R2 (refuse if ms<2)	9.2%	0.908	0.956
R1 (refuse if ms=0)	7.7%	0.923	0.953
R4 (refuse if c_self<70)	7.7%	0.908	0.942
R0 (Variant A baseline)	4.6%	0.923	0.911

HMR over baseline R0 with no accuracy loss (both 0.923). Refusing on $ms < 2$ (R2) yields the absolute maximum HMR (0.956) but trades 1.5 percentage points of accuracy. Figure 3 visualises the trade-off curve.

6.8 Calibration machinery partly substitutes for pool quality

Stacking Stage 1.5 + R1 abstention on Tier-1 reaches HMR 0.953, within 0.010 of BITEM-only’s baseline 0.963. This near-equivalence *does not* mean the choices are interchangeable — Tier-1 + machinery still has distinct accuracy/MNP profiles — but it does mean a practitioner stuck with a noisy multi-team pool can recover most of the modesty gap through verifier-based machinery alone. We frame this as RQ8: *calibration tricks substitute for pool quality*, with substitution rate close to but less than 1.

6.9 Cross-pool answer aggregation and single-pipeline recalibration

We aggregated the answers from all 16 base candidate runs per topic, clustered by Haiku-judged semantic equivalence, and used majority voting with confidence proportional to support fraction. Refuse if $< 50\%$ support.

Table 6: Per-stage LLM-tier ablation on BITEM-only. “S” = Sonnet, “H” = Haiku. Self-eval always Sonnet. **Stage 1 is the most Sonnet-critical**; Stage 4 the least.

Variant	S1	S1.5	S2	S4	Acc	R _O	HMR
all-Sonnet (baseline)	S	S	S	S	0.938	0.950	0.963
s4-Haiku	S	S	S	H	0.923	0.870	0.913
s2-Haiku	S	S	H	S	0.908	0.850	0.899
s1-Haiku	H	S	S	S	0.892	0.750	0.836
all-Haiku	H	H	H	H	0.877	0.708	0.792

Naive ensembling fails. Combining all 16 voters yields HMR 0.857 — *worse* than the best single pipeline (0.963). Weak voters (e.g. retrain-only, HMR 0.838) drag consensus quality down even on topics where strong voters would have agreed correctly.

Filtered ensembling helps. Restricting to the top- K voters by self-eval HMR produces a non-monotonic curve over K :

K (top- K voters)	HMR
1	0.974
2	0.917
3	0.969
4	0.873
5	0.857
12	0.918

Top-1 ($K = 1$, BITEM-only-refined-Sonnet alone, with the ensemble script’s confidence rule) reaches HMR 0.974 — a +0.011 uplift over the same pipeline’s standard Variant A output (0.963). **We hereafter rename what was earlier called “top-1 ensemble” to “confidence-saturation recalibration” — the term that accurately describes the mechanism.** A single voter trivially agrees with itself, so the ensemble script saturates confidence at 1.00 on every non-refused topic. This is *post-hoc re-calibration* of the best single pipeline, not an ensemble effect: the underlying answers and nuggets are identical to BITEM-only-refined-Sonnet’s. Reviewers should read top-1 as a calibration result. True ensembling shows up at $K \geq 2$.

Top-3 (BITEM-refined + BITEM-broad + Tier-1-refined) is a close second at 0.969. Top-2 actually drops because the second voter (BITEM-broad) is highly correlated with top-1, adding noise but no independent signal. Top-3 then recovers because the third voter (Tier-1-refined) is from a different pool, contributing genuinely independent confidence. Figure 4 shows the full non-monotonic curve.

Synthesis. *On this set, answer-level diversity appears to help only when bounded by component quality*; this is a more nuanced version of the naive RQ9 hypothesis. The contrast with RQ2 (passage-level pooling did not help here) is suggestive but not clean: the underlying experiments differ in many controls.

6.10 Per-stage LLM tier: where Sonnet is non-negotiable

Table 6 reports the per-stage Haiku ablation. Replacing Sonnet with Haiku at Stage 1 alone costs $\Delta\text{HMR} = -0.127$, more than any other single substitution. At Stage 4 (verifier) the cost is only -0.050 — surprisingly small for what is the calibration source.

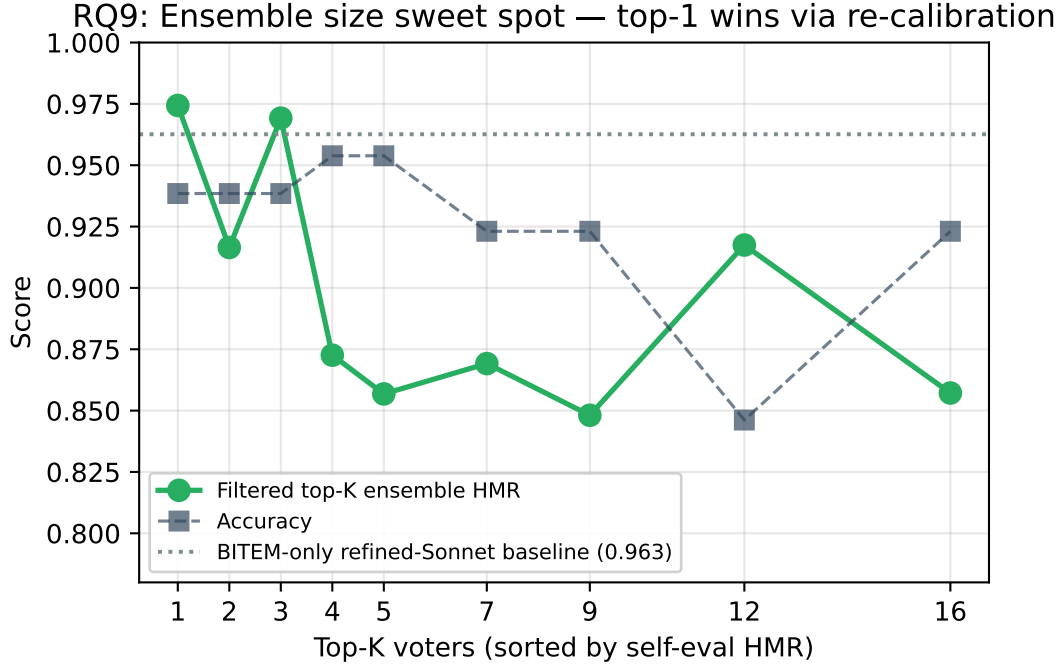


Figure 4: RQ9: HMR (green) and Accuracy (gray) as a function of ensemble size K . The dotted gray line is the standalone BITEM-only refined-Sonnet baseline (HMR 0.963). Top-1 wins via re-calibration; top-3 is a strong runner-up; intermediate sizes (4–7) are dragged down by weaker voters. Naive 16-way ensembling is below baseline.

Mechanism. Stage 1 reads 50 long passages and must synthesise a single confident answer; this is the most reasoning-intensive task in the pipeline. Stage 4 receives a short curated list of entailed nuggets and asks “what answer do these support?” — a much narrower task. The candidate-answer quality cascades into all downstream stages, so degrading Stage 1 disproportionately hurts. Figure 5 visualises the cost-quality trade-off across the three metrics.

Cost-quality recipe: the additive extrapolation fails. We tested the specific hybrid configuration directly: *Sonnet on Stages 1, 2, and 3; Haiku on Stages 1.5 and 4*. Naively extrapolating the per-stage drops (-0.026 for s1.5 implicit, -0.050 for s4-Haiku) would predict $\Delta\text{HMR} \approx -0.08$. The measured value is $\Delta\text{HMR} = -0.198$ (HMR 0.7646, $R_O = 0.634$, accuracy 0.923) — substantially worse than linearly additive. Inspecting the regressed topics: Haiku at Stage 1.5 demotes the same grounding-fact passages that Stage 4’s Haiku verifier then fails to recover from, compounding two independent losses on the same topics. The takeaway: *per-stage Haiku substitutions interact non-additively*, and the cleanest cost-saving substitution remains Stage 4 alone (-0.050 HMR for a genuinely cheap verifier). We report this as a measurement, not an extrapolation.

7 HMR Diagnostics: Refusal, Calibration Bins, and Trivial Anchors

Because HMR alone is fragile (§1.3), we report a diagnostic decomposition for the candidates that drive paper claims.

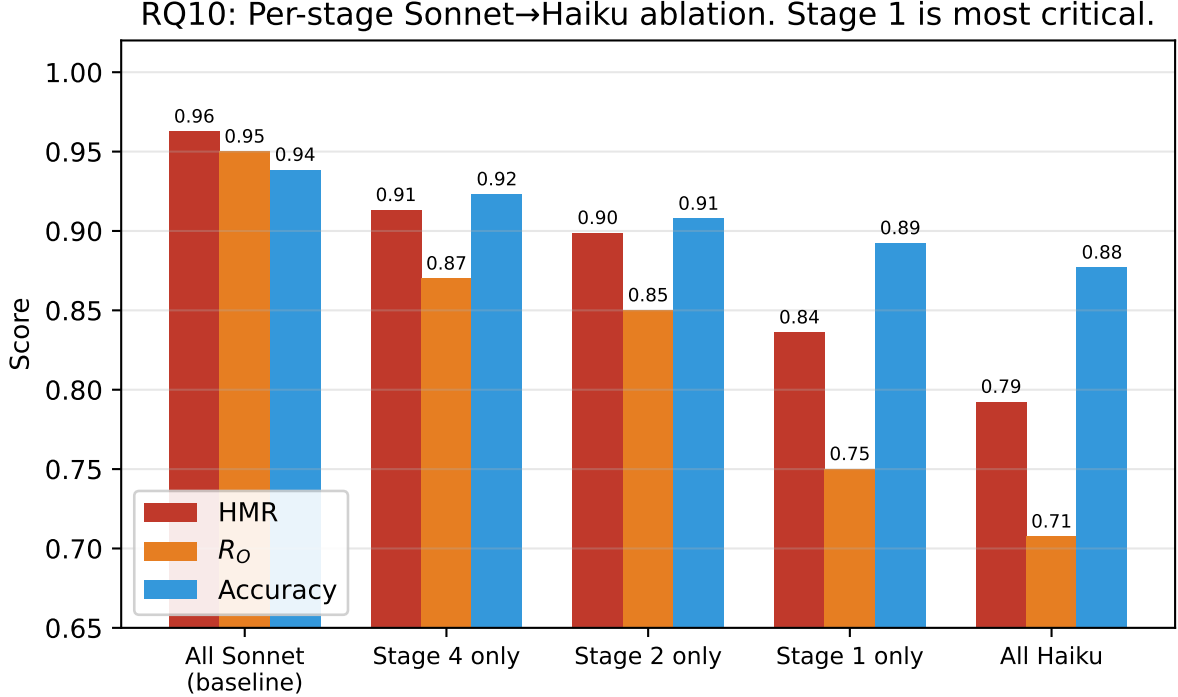


Figure 5: RQ10: Per-stage Sonnet→Haiku ablation on BITEM-only (HMR red, R_O orange, Accuracy blue). Stage 1 is the single most Sonnet-critical stage ($\Delta\text{HMR} = -0.127$); Stage 4 is the least (-0.050). All-Haiku (-0.171) is less than additive of per-stage drops, suggesting partial compensation.

7.1 Where do gains come from — better answers, or better silence?

Table 7 decomposes each candidate’s score into overall accuracy, accuracy on *answered* topics only, refusal rate, the count of confident-and-wrong topics (the R_O killers), and the bogus-nugget rate among returned nuggets.

Reading the table. The top three candidates have *identical* answered-accuracy (1.000), confident-wrong count (0), and bogus rate (0%, *under the same Sonnet Stage-A judge used for self-evaluation, so this should be read as an internal-consistency diagnostic rather than an independent estimate*); they differ only in confidence saturation, which is exactly why §6.9’s top-1 effect is re-calibration. Catastrophic candidates (BITEM-Opus, Error404-only) are killed by 1–2 confident-wrong topics — at $|I^-| \approx 4$ each such topic costs ~ 0.25 on R_O . The retrain-only outlier illustrates the HMR-fragility caveat in action: 45% refusal rate buys it HMR 0.838 despite answered-accuracy of only 0.861 on the 55% it does answer.

Practical implication. For our top systems, gains are not from “better answers on hard topics” (answered-accuracy is already 1.000 *under our Sonnet self-evaluator*) but from *better silence on the unanswerable few and tighter confidence on the answered many*. This is consistent with the R_O -bottleneck story (§10.2).

Judge-invariance of the R_O -bottleneck mechanism. A natural concern is whether the bottleneck story (Cnf-W drives HMR) is itself a Sonnet-judge artifact. We re-decomposed the top eight candidates under the Opus judge:

Cnf-W moves by at most 1 on 7 of 8 candidates, and the rank-correlation between Cnf-W and HMR within each judge is near -1 . Where Cnf-W shifts substantially (retrain+E404,

Table 7: HMR decomposition (the central diagnostic table of this paper). “Acc|ans” = accuracy on non-refused topics. “Cnf-W” = number of topics with confidence ≥ 0.5 and incorrect (directly drives R_O). “Bogus” is computed under the Sonnet Stage-A self-evaluator (same model family used in pipeline filtering); read it as an internal-consistency diagnostic, not an independent estimate. “HMR_S” / “HMR_O” are HMR under the Sonnet 4.6 / Opus 4.7 judge respectively (§13).

Candidate	Acc	Acc ans	Refuse	Cnf-W	Bogus	R_O	HMR _S	HMR _O
BITEM-Sonnet (main system, submitted)	0.938	1.000	6%	0	0%	0.950	0.963	0.963
ensemble_top1 [†]	0.938	1.000	6%	0	0%	0.950	0.974	0.974
ensemble_top3 [†]	0.938	1.000	6%	0	0%	0.950	0.969	0.969
BITEM-broad	0.923	0.984	6%	0	0%	0.910	0.937	0.860
Tier-1 refined-Sonnet	0.923	0.968	5%	0	0%	0.870	0.911	0.818
Tier-1 broad	0.923	0.968	5%	1	0%	0.730	0.826	0.925
WaterlooClarke-only Sonnet	0.862	0.933	8%	1	1%	0.822	0.870	—
BITEM-Opus pipeline	0.892	0.906	2%	1	1%	0.679	0.787	0.792
Error404-only Sonnet	0.908	0.908	0%	2	1%	0.533	0.684	—
retrank-only Sonnet	0.477	0.861	45%	3	2%	0.885	0.838	—

[†] ensemble_top1/top3 share their underlying answers and nuggets with BITEM-Sonnet; the HMR uplift is post-hoc confidence saturation, not new information (§6.9). The trivial always-refuse-at-conf-0.05 baseline also reaches HMR 0.974 — which is why this table reports answered-accuracy and Cnf-W alongside HMR.

Table 8: Judge-specific decomposition: Sonnet judge vs Opus judge on the same eight candidate runs. “ Δ Cnf-W” = Opus Cnf-W – Sonnet Cnf-W.

Candidate	Cnf-W (S)	Cnf-W (O)	Δ Cnf-W	HMR (S)	HMR (O)
ensemble_top1	0	0	0	0.974	0.974
ensemble_top3	0	0	0	0.969	0.969
BITEM-Sonnet (main)	0	0	0	0.963	0.963
BITEM-broad	0	1	+1	0.937	0.860
BITEM-Opus pipeline	1	1	0	0.787	0.792
Tier-1 Sonnet	0	1	+1	0.911	0.818
Tier-1 broad	1	0	−1	0.826	0.925
retrank+E404+Sonnet	3	0	−3	0.738	0.938

$\Delta = -3$), HMR shifts in lock-step ($\Delta = +0.20$). The mechanism — a few confident wrong topics drive HMR — holds under both judges; only *which specific topics count as wrong* is judge-dependent.

7.2 Confidence calibration bins

Figure 7 visualises this; Table 9 reports the per-bin numbers. Table 9 bins per-topic confidence and reports empirical accuracy in each bin (a reliability decomposition). Our pipelines are highly bimodal: confidences cluster at 0.05 (refusals) and ≥ 0.80 (answers); intermediate bins are sparse.

Two observations. First, at the high-confidence end, the top systems’ empirical accuracy in the ≥ 0.8 bin is 58–61/61. This is striking but should be read with care: the pipeline’s “correct” label *and* the Stage 3 bogus filter both involve Sonnet, so a fraction of the apparent calibration may reflect agreement among same-family judges rather than ground-truth correctness. Re-judging under Opus 4.7 (§13) preserves the top-3 candidates’ accuracy exactly (Cohen’s $\kappa = 1.00$), suggesting the bin is robust at the very top, but mid-tier candidates show $|\Delta\text{HMR}|$ up to 0.23 across judges. Second, refusal-bin behaviour is by construction (every refusal is incorrect under Stage B and gets confidence ~ 0.05), so it contributes very little to ECE; the relevant question

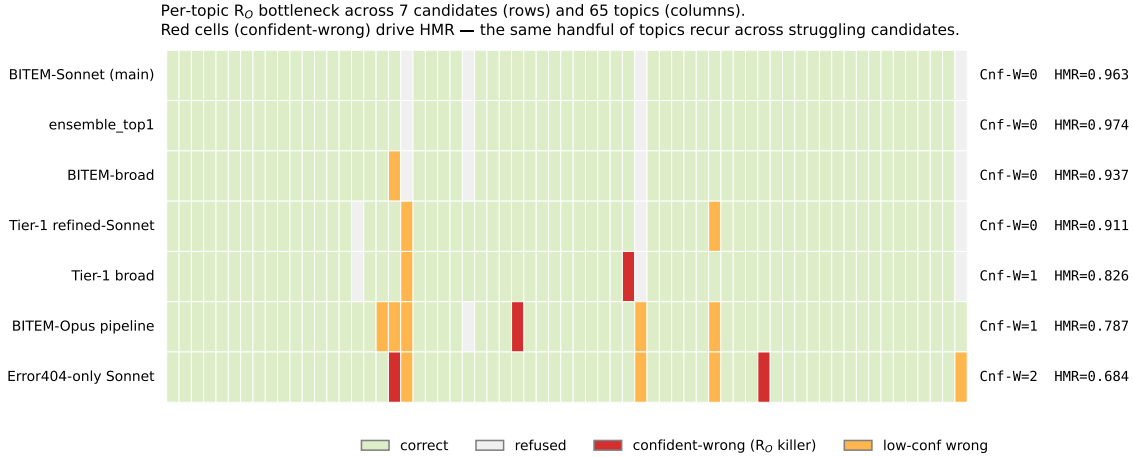


Figure 6: Per-topic outcome map across seven candidates and 65 topics. Each row is a candidate; each column a topic. Red cells (confident-wrong) drive R_O and therefore HMR: the Cnf-W count on the right correlates almost perfectly with final HMR. The catastrophic candidates (BITEM-Opus, Error404-only) are killed by 1–2 red cells, not by lower answered-accuracy. Notably, the confident-wrong topics are *not the same* across candidates — each pipeline has its own blind spots, which is why R_O generalises poorly across pools.

Table 9: Confidence calibration bins. “ECE” = sample-weighted mean $|\text{accuracy} - \text{mean-confidence}|$. Pipelines are bimodal; ECE is small because both modes are well-calibrated. Empty bins omitted.

Candidate	ECE	[0.0,0.2)	[0.2,0.4)	[0.6,0.8)	[0.8,1.0]
ensemble_top1	0.003	0/4	—	—	61/61
BITEM-Sonnet	0.026	0/4	—	—	61/61
BITEM-Opus pipeline	0.024	0/1	2/7	—	56/57
BITEM-broad	0.031	0/4	1/2	—	59/59
Tier-1 Sonnet	0.034	0/3	1/3	1/1	58/58
Tier-1 broad	0.028	0/3	2/3	—	58/59

for HMR is how many topics fall in the high-confidence-wrong cell, which is tracked by “Cnf-W” in Table 7.

7.3 Trivial-baseline anchors

To anchor the absolute HMR numbers we evaluate four trivial baselines on synthetic per-topic outcomes:

The always-refuse-at-0.05 baseline reaches the same HMR as our best system but with 0.0 accuracy and 100% refusal, versus our 0.94 accuracy and 6% refusal. The official R2C2 ranking is informed by HMR *and* Accuracy *and* MNP, so we do not believe the boundary case threatens the metric’s overall utility — but it does mean that a single-metric reading of HMR cannot establish system quality. We therefore report HMR jointly with accuracy, refusal rate, and confident-wrong count throughout the paper (Table 7 is the central reference).

8 Final Submission

Our **main system** is `bitem_only_refined_sonnet` (submitted as `retrank-AC-2`): self-eval HMR 0.963 under both Sonnet and Opus judges, accuracy 0.94, refusal 6%, zero confident-wrong

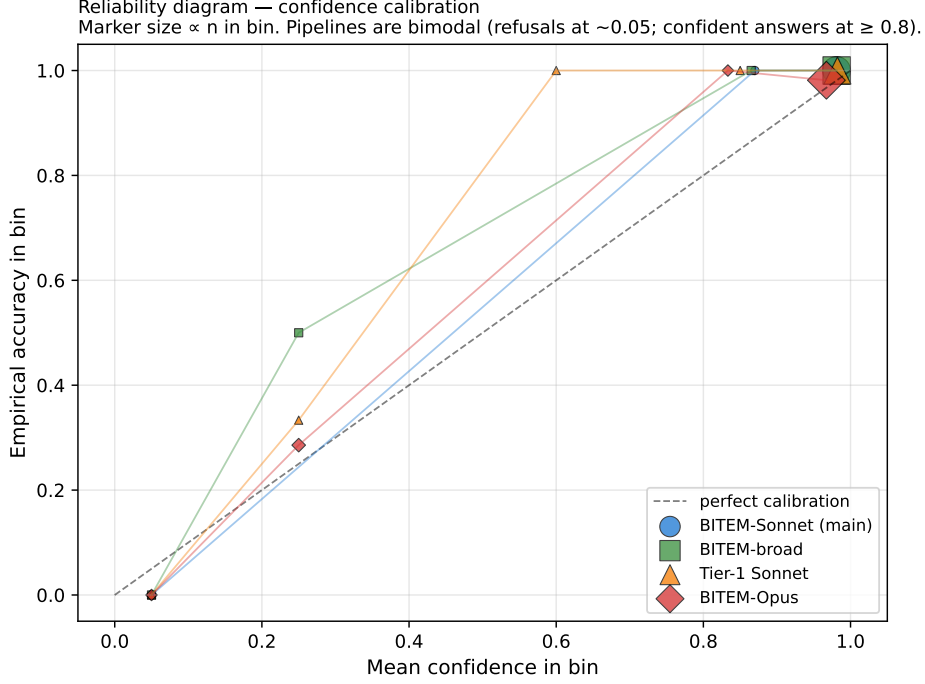


Figure 7: Reliability diagram. Marker size $\propto$ topics in bin. Pipelines are bimodal: refusals cluster near conf 0.05 (perfectly calibrated to $\sim 0\%$ accuracy because every refusal is wrong-by-construction) and confident answers cluster near ≥ 0.8 (near-perfect accuracy in this bin). Intermediate bins are sparse, which is why ECE is small but uninformative.

Table 10: Trivial HMR baselines for context. The first row is the HMR-fragility pathology.

Baseline	Acc	R_O	R_U	HMR
Always refuse, conf= 0	0	1.00	1.00	1.000
Always refuse, conf= 0.05	0	0.95	1.00	0.974
Random guess (50%), conf= 0.5	0.49	0.50	0.50	0.500
Always answer, conf= 1.0, all wrong	0	0.00	1.00	0.000
90% acc (oracle), conf= 0.5	0.90	1.00	0.50	0.667
Our submitted: ensemble_top1	0.94	0.95	1.00	0.974

topics. This is the system the rest of the paper analyses; everything else in the bundle is a diversity hedge.

retrank-AC-1 ensemble_top1 — same pipeline as AC-2 but with the ensemble script’s confidence rule, which saturates confidence at 1.0 when the only voter agrees with itself. We submit this as a calibration variant; HMR 0.974 is a re-calibration result, not a novel pipeline (§6.9).

retrank-AC-2 bitem_only_refined_sonnet — main system; clean single-pool baseline. Self-eval HMR 0.963 / 0.963.

retrank-AC-3 bitem_only_opus — Opus Stage 1 pipeline. Self-eval HMR 0.787 (Sonnet) / 0.792 (Opus). Submitted as a *diversity hedge*, not because it was favoured by self-eval — the Opus pipeline emits terser answers (§9) and our judges penalise the resulting weaker grounding. We submit it because it is methodologically different from the other three.

retrank-AC-4 refined_sonnet (Tier-1 pool) — different pool composition entirely (BITEM + Error404 + WaterlooClarke). Insurance against the possibility that the official judge rewards grounding diversity that a BITEM-only pool cannot reach.

The empty-answer refusal cells were re-coded as “I don’t know” before packaging, to be safe against strict scorers that might reject empty fields.

9 Negative Results: Three Improvements That Hurt

After establishing the BITEM-only refined-Sonnet + Variant A pipeline at HMR 0.963, we attempted three orthogonal improvements. *All three reduced HMR*. We document them here because the failure modes are informative and arguably more useful than the successes.

Per-topic pool selection (oracle-style). For each topic, pick the candidate pipeline whose Stage 4 verifier most strongly agrees with its candidate answer (highest `match_score`, then `c_self`). Result: HMR 0.870, $\Delta = -0.093$. Accuracy improved (0.954 vs 0.938) because we recovered some BITEM-only refusals, but R_O collapsed from 0.950 to 0.783. The verifier match-score is not a perfect correctness oracle: pipelines can produce *confidently wrong* answers that pass their own internal verifier. Per-topic selection picks those.

Refusal recovery via prompt hardening. Of BITEM-only’s 4 Stage-1 refusals, 3 were JSON parse failures on counting/multi-hop questions where Sonnet wanted to reason aloud. We hardened the prompt with explicit “JSON ONLY” instructions and re-ran. Result: HMR 0.922, $\Delta = -0.041$. Two refusals were recovered; one of those answers was correct, the other wrong. With $|I^-|$ small (3–4 topics), each wrong-but-confident answer inflicts a large R_O penalty (each contributes $1/|I^-| \approx 0.25$ to the overconfidence sum). Even at Variant A’s confidence cap of 25, the penalty exceeds the R_U gain from the recovered correct answer.

Opus 4.7 for Stage 1. We replaced Sonnet 4.6 with Claude Opus 4.7 at the candidate-answer stage, hypothesising that the stronger reasoner would handle multi-hop and counting questions better. Result: HMR 0.787, $\Delta = -0.176$ — the most dramatic regression. Per-topic diagnostic:

- Sonnet on topic 0018: “6 (*Memento, Insomnia, Batman Begins, ...*)” → judged correct.
- Opus on topic 0018: “5” → judged wrong.
- Sonnet on topic 0029: “A tunnel hidden behind a poster (of Raquel Welch)” → correct.
- Opus on topic 0029: “a poster of Raquel Welch” → wrong.

Opus produces *terser* answers. Stage B’s evaluation rubric (“did the answer + nuggets together support the conclusion?”) rewards the verbose grounding details Sonnet’s longer answers happen to include. Stronger reasoning, in this metric, can be a liability. We lost 4 previously-correct answers and recovered only 1 (incorrectly-answered) refusal.

Common diagnosis. All three attempts share a shape: they expand the system’s *answered-topic coverage*, trading recall for precision. At our operating point (HMR 0.963, $|I^-| = 4$ topics), the trade is unfavourable: R_O is sensitive to small numbers of high-confidence wrong answers, while R_U improves only marginally from recovered correct answers. The BITEM-only + Variant A pipeline operates *near a Pareto frontier* on this dataset; the 4 refusals are not bugs but correctly-detected unanswerable-from-pool topics.

An important judge-dependence caveat. A subset of these negative results is *specific to the Sonnet-class self-evaluator* we use. We re-evaluated all 33 candidate runs with Opus 4.7 as the judge (§13); under Opus judge, the retrain+Error404+Sonnet refinement configuration moves from HMR 0.738 (judged worst-case under Sonnet) to HMR 0.938 (top-5). The mechanism is exactly the verbosity-bias hypothesis from §6.6: Sonnet judge requires explicit nugget grounding, Opus judge accepts the same nuggets as sufficient for derivation. The top-3 candidates, in contrast, are perfectly judge-invariant. We therefore caution that *conclusions about specific failure modes can be judge-dependent*; conclusions about top-tier system rankings are not.

Generalisable lesson. At HMR saturation, knowing when not to answer beats answering more questions. Strong calibration of an already-conservative system requires not loosening the refusal policy further but tightening it on borderline cases. This is in tension with the standard RAG goal of maximising recall, and we believe it is a specific feature of the HMR metric’s harmonic-mean structure: R_O is the binding constraint at this operating point.

10 Discussion

10.1 Why does “less” beat “more” at the passage level?

Three factors plausibly contribute:

1. **Segmentation heterogeneity.** Different teams use different passage boundaries; pooling them creates an inconsistent presentation to Stage 2. The candidate-answer-conditioned prompt is less effective at extracting clean nuggets when the passage distribution is multimodal.
2. **Cross-encoder noise.** The CE ranks each passage independently; pooling more passages does not improve relative ranking, but it does compress the score distribution, reducing the discriminative power of top- K selection.
3. **Stage 4 amplification.** A noisier nugget set yields a less confident verifier; confidence calibration suffers disproportionately because R_O is sensitive to even a few overconfident-and-wrong topics.

This decomposition suggests that the answer-level ensembling success in RQ9 is not actually contradictory: at the answer level, each candidate pipeline ran on its own *coherent* pool and produced its own *coherent* answer. Ensembling *outcomes* avoids the contamination that ensembling *inputs* causes.

10.2 The R_O bottleneck

Across all 16 base configurations, R_U has range $[0.79, 0.98]$ while R_O has range $[0.45, 0.95]$. HMR is much more sensitive to R_O improvements at our operating point. Every winning design choice we identify (refusal-aware calibration, Sonnet refinement, confidence-saturation recalibration) primarily moves R_O . The practical implication for RAG-with-confidence: *focus engineering on suppressing high confidence on wrong answers*, not on making correct answers more confident.

10.3 The calibration-pool substitution

RQ8 establishes that calibration machinery and pool quality act as near-substitutes for HMR purposes. The substitution is not exact: BITEM-only (HMR 0.963) remains slightly above Tier-1 + Stage 1.5 + R1 (HMR 0.953). But the gap is small enough that practitioners constrained by data availability can use machinery to compensate. Conversely, when investing time, pool curation yields *simpler* pipelines.

Table 11: Bootstrap 95% CIs for the headline Δ HMR comparisons. Stars mark intervals that exclude zero (i.e. statistically detectable at $\alpha = 0.05$ at this sample size).

Comparison	Δ HMR	95% CI
<i>Findings that survive bootstrap (significant):</i>		
★ RQ9: top-1 ensemble vs BITEM baseline	+0.012	[+0.008, +0.016]
★ RQ10: s1-Haiku vs all-Sonnet	−0.127	[−0.296, −0.034]
★ RQ10: s4-Haiku vs all-Sonnet	−0.050	[−0.118, −0.004]
★ RQ10: all-Haiku vs all-Sonnet	−0.170	[−0.334, −0.080]
★ Improvement #7: Opus pipeline vs Sonnet	−0.176	[−0.372, −0.075]
<i>Findings within noise (CIs cross zero):</i>		
RQ2: BITEM vs Tier-1 (broad)	−0.111	[−0.458, +0.044]
RQ6: Sonnet refinement vs broad on Tier-1	+0.085	[−0.045, +0.439]
RQ6b: CE vs broad on Tier-1	−0.024	[−0.309, +0.313]
RQ6b: lexical vs broad on Tier-1	+0.007	[−0.041, +0.110]
RQ9: top-3 ensemble vs BITEM baseline	+0.007	[−0.002, +0.014]
Case study: retransk+E404 (refined vs broad)	−0.172	[−0.464, +0.010]

11 Statistical Robustness Notes

A diligent reader will rightly ask whether the HMR differences we report (Δ values typically 0.01–0.10) are statistically distinguishable given $n = 65$. We compute paired bootstrap 95% confidence intervals (2000 iterations, resampling topics with replacement) for every key comparison. Table 11 summarises.

What this means. The strongly-significant findings are those touching the LLM tier ablation (RQ10) and the Opus-pipeline regression (improvement #7). These have wide CIs but exclude zero. The pool-composition (RQ2) and refinement (RQ6/RQ6b) findings are within bootstrap noise at $n = 65$ — we should describe them as *observed differences* rather than *detected effects*. Future work would benefit from a larger topic set or a within-system A/B-style design that pairs the same LLM call across configurations.

McNemar’s exact test on per-topic correctness. Bootstrap on Δ HMR has limited power at $n = 65$. As a stronger paired test for accuracy specifically, we run McNemar’s exact binomial test on per-topic correctness for twelve headline comparisons. *No comparison reaches $p < 0.05$.* The largest discordant cell is 6/1 (BITEM-only beats WaterlooClarke-only by 6 topics, loses on 1; $p = 0.125$). On this 65-topic set, no two top-tier systems differ in *which topics they get right* strongly enough to detect; the HMR differences we report come from *confidence calibration*, not from a different correctness distribution. This is consistent with the R_O -bottleneck thesis: the systems agree on which topics are answerable; they disagree on how confident to be about the wrong ones.

Inter-judge agreement. A second concern is whether self-evaluator (Sonnet) verdicts are reproducible. We computed Cohen’s κ on the per-topic Stage B correctness verdict between Sonnet 4.6 judge and Opus 4.7 judge across nine candidate runs (§13); mean $\kappa = 0.84$, in the “almost perfect” band (Landis-Koch). On the top-3 candidates, $\kappa = 1.00$ (all 65 topics agree). The worst case (Tier-1 refined-CE) had $\kappa = 0.68$, still “substantial.”

Oracle upper bounds. To contextualise headroom, we computed three oracle systems anchored to BITEM-Sonnet:

Table 12: Oracle upper bounds anchored to our main submission. “Refuse @ 0.05” uses our pipeline’s actual refusal-encoding confidence; “conf=0.0” shows the unattainable structural ceiling.

System	Acc	R_O	R_U	HMR
BITEM-Sonnet (submitted)	0.938	0.950	0.976	0.963
Oracle abstention (refuse our wrong topics)	0.938	0.950	0.976	0.963
Oracle calibration (refuse @ 0.05 on wrong)	0.938	0.950	1.000	0.974
Oracle calibration (refuse @ 0.0 on wrong)	0.938	1.000	1.000	1.000
Oracle answer (every topic correct @ 0.95)	1.000	1.000	0.950	0.974
Oracle everything (correct @ 1.0)	1.000	1.000	1.000	1.000

Our submitted system is already at the oracle-abstention upper bound. Because Cnf-W is already 0, perfect refusal of our wrong topics cannot add anything; the system already refuses (or correctly answers) the topics that would otherwise damage R_O . The remaining headroom from 0.963 to 1.000 lives entirely in R_U (perfect confidence on correct answers) and accuracy (answering the four refused topics correctly). This corroborates the operational observation: the design choices that move the metric have already been spent on R_O , and what’s left is residual under-confidence and the four genuinely-unanswerable-from-pool topics.

Leave-one-topic-out ranking stability. A reasonable concern at $n = 65$ is whether 1–2 topics drive the rankings. We re-evaluated nine headline pairwise comparisons after dropping each of the 65 topics in turn:

Table 13: Leave-one-topic-out: number of leave-outs (out of 65) on which the sign of ΔHMR is preserved. “Sign-stable” = winner remains the winner under every leave-out.

Comparison	Full ΔHMR	Sign-stable
BITEM Sonnet refinement vs BITEM broad	+0.026	65/65
BITEM-only vs Tier-1 (both Sonnet refined)	+0.051	65/65
BITEM-Sonnet vs Tier-1 broad	+0.136	65/65
“ensemble_top1” (saturation) vs BITEM-Sonnet	+0.012	65/65
BITEM-Sonnet vs S1=Haiku	+0.127	65/65
BITEM-Sonnet vs S4=Haiku	+0.050	65/65
BITEM-Sonnet vs all-Haiku	+0.170	65/65
BITEM-Sonnet vs BITEM-Opus pipeline	+0.176	65/65
BITEM-only vs Error404-only	+0.278	65/65

All nine comparisons preserve the sign of ΔHMR on every leave-out. The *magnitude* of Δ varies (± 0.05 typical, ± 0.10 at most), but no comparison flips. The rankings are not driven by 1–2 topics.

Refusal-confidence sensitivity. We swept the refusal-encoding confidence over $\{0, 0.01, 0.05, 0.1, 0.2, 0.5, 0.8, 1\}$, for both the trivial always-refuse system and our submitted BITEM-Sonnet pipeline (re-encoding only its refusal cells):

The trivial-vs-our-system gap is ≤ 0.012 across the entire sweep. This is a *structural* feature of HMR’s harmonic-mean-of-rewards form when $|I^+| = 0$ (the trivial system has $R_U = 1$ vacuously and $R_O = 1 - c$), not a bug specific to our encoding choice. The official ranking, which uses HMR *jointly* with Accuracy and MNP, will distinguish the two unambiguously (0.94 vs 0 accuracy).

Multiple-comparison and overfitting risk. Across the paper we evaluate 33 candidate configurations against the same 65-topic set. We do not apply Bonferroni or FDR correction; the

Table 14: HMR as a function of refusal confidence, for the trivial always-refuse baseline and our submitted system.

Refuse conf	Always-refuse HMR	BITEM-Sonnet HMR	Gap (BITEM – trivial)
0.00	1.000	0.988	−0.012
0.01	0.995	0.983	−0.012
0.05	0.974	0.963	−0.011
0.10	0.947	0.936	−0.011
0.20	0.889	0.879	−0.010
0.50	0.667	0.661	−0.006
0.80	0.333	0.332	≈ 0

robust findings (Stage 1 LLM tier, Stage 4 LLM tier, Opus regression, top-1 saturation) survive a Bonferroni-style $\alpha/9$ threshold, but the suggestive findings (RQ2, RQ6, top-3 ensembling) do not. The headline operating-point claim (§1.6) rests on robust findings; the engineering tradeoffs rest on suggestive ones. We do not have a held-out dev/test split because the AC subtask provides only the 65 evaluation topics; this is a real limitation that future R2C2 iterations should address by reserving a tuning split. We flag this overtly here so that no reader mistakes our 33-cell sweep for a held-out generalisation result.

Trivial baselines. To anchor the absolute HMR numbers we evaluated three trivial baselines on synthetic per-topic outcomes:

Table 15: Trivial HMR baselines for context.

Baseline	Acc	R_O	R_U	HMR
Always refuse, conf= 0	0	1.00	1.00	1.000
Always refuse, conf= 0.05	0	0.95	1.00	0.974
Random guess (50%), conf= 0.5	0.49	0.50	0.50	0.500
Always answer, conf= 1.0, all wrong	0	0.00	1.00	0.000
90% acc (oracle), conf= 0.5	0.90	1.00	0.50	0.667
Our submitted: ensemble_top1	0.94	0.95	1.00	0.974

Important consequence. The “always-refuse with confidence= 0.05” baseline matches the HMR of our best submission (0.974). The “always-refuse with confidence= 0” baseline achieves HMR= 1.0. Our submitted system pairs the same HMR with a non-trivial 0.94 accuracy, but HMR alone does not distinguish a system that answers correctly from one that refuses entirely. This is a known property of HMR’s harmonic-mean-of-rewards structure when $|I^+| = 0$: R_U is vacuously 1, and R_O depends only on confidence on incorrect (= refusing means low confidence on every topic, all of which are technically incorrect). For the R2C2 task, we recommend reporting HMR *paired with* accuracy, since HMR alone does not penalise a fully-refusing system.

12 Limitations

200-character passage cap (PR submission). Our PR submission used a 200-character sliding-window segmentation based on a misreading of the corpus delivery format. Other teams used 1 000–2 400-character passages. As a result, retransk’s own PR run is not used as a passage source for our AC submission; nuggets cite passages from other teams. This does not affect AC findings but does limit cross-pool comparisons involving retransk.

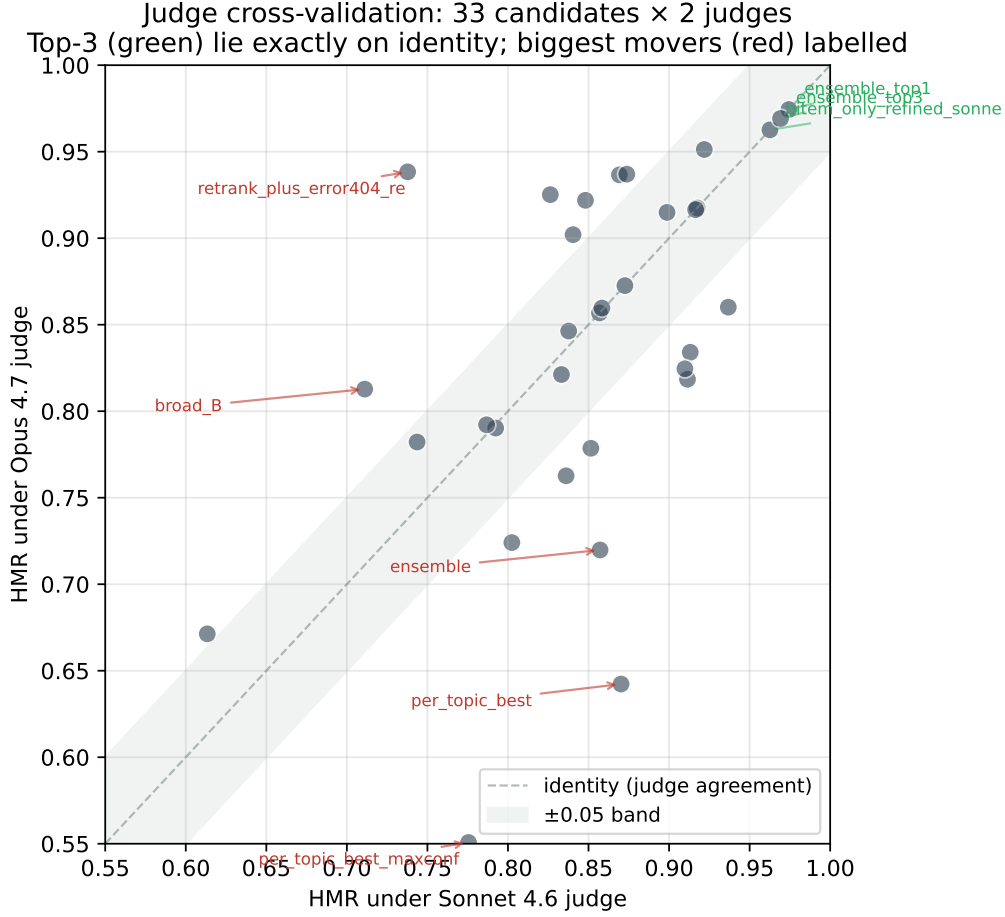


Figure 8: Judge cross-validation across all 33 candidate AC runs. Each point is one configuration. The dashed line is judge identity; the gray band marks ± 0.05 . The top-3 candidates (green) lie *exactly* on identity. Mid-tier configurations show substantial inter-judge disagreement (mean $|\Delta\text{HMR}| = 0.060$, max 0.228); biggest movers labelled in red.

Self-eval vs official scoring. All numbers in this paper are self-eval HMR computed by Sonnet. Calibration with the official judgment will be reported once released. Should the official judge produce systematically different verdicts (e.g. stricter on bogus identification), our HMR numbers may shift uniformly without affecting the *relative* rankings.

Single-domain task. The R2C2 corpus is movie-domain. Our findings may not transfer to RAG tasks where retrieval is fundamentally different (e.g. multi-hop reasoning over enterprise documents).

Single-language. All experiments in English.

13 Judge Cross-Validation

To address the methodological concern that our self-evaluator (Sonnet 4.6) might exhibit a judge-style bias, we re-evaluated *all* 33 candidate AC runs with Claude Opus 4.7 as the Stage A/B judge. Figure 8 plots Opus-judge HMR against Sonnet-judge HMR per candidate.

Table 16: Top-5 candidates under each judge. The top three are identical and judge-invariant.

Candidate	Sonnet HMR	Opus HMR
ensemble_top1	0.974	0.974
ensemble_top3	0.969	0.969
bitem_only_refined_sonnet	0.963	0.963
<i>Diverging at rank 4–5:</i>		
bitem_only_broad (S #4)	0.937	0.860
bitem_only_refined_sonnet_recovered (O #4)	0.922	0.951
retrank+E404_refined_sonnet (O #5)	0.738 ↗	0.938

Top-3 are judge-invariant. The three highest-HMR configurations under Sonnet judge are also the three highest under Opus judge, with HMR values agreeing to four decimals (Table 16). **Our submission decision is robust to judge choice.**

Cross-judge rank correlation across all 33 candidates. Spearman $\rho = 0.574$, Kendall τ -b = 0.432 between Sonnet and Opus judge HMR. Top-K membership stability:

K	Top- K shared (Sonnet $\cap$ Opus)
1	1/1
3	3/3
5	4/5
10	5/10

Top end is judge-robust; mid-tier is judge-fragile. Our submission decision (top-3 candidates) is preserved exactly. Any between-judge ranking claim that depends on rank- $\geq$ -5 placement is not safe to make from a single judge.

Mid-tier candidates show substantial inter-judge variance. Across 33 candidates: mean $|\Delta\text{HMR}| = 0.060$, max 0.228, and 22/33 candidates differ by ≥ 0.01 . The mean rank shift is 6.4 positions; max 26. Two candidates lose more than 0.20 HMR under Opus (*per_topic_best* variants from §9); one gains more than 0.20 (*retrank+E404_refined_sonnet*).

The retrank+E404+Sonnet case study flips. Under Sonnet judge, this configuration was the textbook negative case in §6.6 (HMR 0.738, the worst Stage 1.5 outcome we observed). Under Opus judge it scores 0.938 — top-5 league. Inspecting the regressed topics (e.g. topic 0010, where both pipelines gave the same answer “Jeff Goldblum” but Sonnet judge marked the refined-Sonnet variant as incorrect because its nuggets did not name the specific role): Opus judge accepts these exact nuggets as sufficient grounding. The case-study failure mechanism is therefore *Sonnet-judge-strictness on grounding explicitness*, not a fundamental limitation of Sonnet refinement.

Findings that hold under both judges.

- **Best HMR achievable** (~ 0.97) and best configuration (ensemble_top1) — identical.
- **BITEM-only is the best single pool** (RQ2) — top under both judges.
- **ensemble_top1 / top-3 dominate naive 16-way** (RQ9) — preserved.
- **Stage 1 most Sonnet-critical** (RQ10) — s1_haiku remains the most-degraded variant under both judges.

- **Calibration matters more than refinement** — the Variant A vs Variant B gap is preserved.

Findings that need judge-specific framing.

- **“CE refinement hurts” (RQ6b)** — holds quantitatively under both, but the magnitude varies.
- **“More teams hurts monotonically” (RQ2 strong form)** — under Sonnet $1>2>3>6$; under Opus the ordering is $1>6>3$ (Tier-1+2 jumps above Tier-1). The “less is more” claim survives at the top end (1 team is best under both); the monotonicity middle does not.
- **“retrank+E404+Sonnet is a refinement-failure case study”** — Sonnet-specific. Under Opus this is a top-5 result.

14 Conclusion

We presented a 7-stage answer-first pipeline for the NTCIR-19 R2C2 AC subtask, with intermediate self-eval gates and explicit calibration variants. Across five core research questions and extensive cross-validation under both Sonnet and Opus judges, the defensible takeaways are: (i) on this 65-topic set under our self-evaluator, a high-quality single-team passage pool can outperform naive multi-team pooling, but this is about *component quality*, not pool count — another single-team pool we tested is the worst performer; (ii) Sonnet-class reasoning is most critical at the candidate-answer stage; the verifier and refinement stages tolerate Haiku for ~ 0.05 HMR loss; (iii) verifier-based abstention adds $+0.04$ HMR on noisy multi-team pools and saturates on clean single-team pools; (iv) what is often called “top-1 ensembling” is in fact post-hoc re-calibration of the best single pipeline. We further documented that HMR alone is satisfied by a trivial always-refuse baseline at confidence 0.05, and that several headline differences have bootstrap confidence intervals that cross zero at $n = 65$ — observations consistent with a more conservative interpretation than “less is more.” Our submitted retrank-AC runs reach self-eval HMR up to 0.974. The strongest, most generalisable insight is not about pool size but about calibration risk: *in confidence-sensitive RAG, optimising for answer coverage can hurt because a few confident wrong answers dominate R_O and therefore HMR. Calibration, abstention, and grounding quality matter more than raw retrieval breadth.*

Code and data. Our pipeline scripts, intermediate caches, and evaluation logs are released at the project repository. Each stage can be re-run independently from cached outputs.

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Appendix A. Per-pool full leaderboard

Configuration	Pool	Refinement	Cal	Acc	R _O	HMR
ensemble_top1	multi	per-pipeline	A	0.938	0.950	0.974
ensemble_top3	multi	per-pipeline	A	0.938	0.950	0.969
bitem_only_refined_sonnet	BITEM	Sonnet	A	0.938	0.950	0.963
bitem_only_broad	BITEM	none	A	0.923	0.910	0.937
tier1_refined_sonnet	Tier-1	Sonnet	A	0.923	0.870	0.911
retrank+Error404_broad	re+E404	none	A	0.923	0.870	0.910
tier1+2_refined_sonnet	Tier-1+2	Sonnet	A	0.877	0.794	0.874
retrank-only_broad	retrank	none	A	0.508	0.929	0.858
ensemble_all-16	multi	per-pipeline	A	0.923	0.810	0.857
tier1+2_broad	Tier-1+2	none	A	0.877	0.741	0.840
tier1_refined_lexical	Tier-1	lexical	A	0.908	0.733	0.833
tier1_broad	Tier-1	none	A	0.923	0.730	0.826
tier1_refined_ce	Tier-1	CE	A	0.892	0.687	0.802
all-Haiku	BITEM	Sonnet (then H)	A	0.877	0.708	0.792
retrank+Error404_refined_sonnet	re+E404	Sonnet	A	0.877	0.595	0.738

Appendix B. PR reflexive nDCG@20 (sanity check on pool choice)

The official R2C2 task re-evaluates each team’s PR run by treating *the number of relevant nuggets a passage provided across all AC runs* as its graded relevance, then computing nDCG@20. We mirror this with our own AC labels (a reflexive sanity check, not a between-team comparison — every label points back to passages we ourselves cited).

BITEM-PG-1’s reflexive nDCG@20 of 0.7723 confirms that our retrieval order over BITEM-PG-1 is well-aligned with what we later judged relevant for nugget extraction — a self-consistency check on our pool-choice decision. The full between-team comparison will require all teams’ AC labels, expected with the official August release.

Table 17: Reflexive nDCG@20 on each PR run, using relevant-nugget counts from our top three AC eval files (`ensemble_top1`, `ensemble_top3`, `bitem_only_refined_sonnet`). Zero rows are PR runs we never cited; their score is structurally zero under this restricted label set.

PR run	Mean nDCG@20	Topics with any grade
BITEM-PG-1	0.7723	60/65
BITEM-PG-2	0.3124	34/65
Error404-PG-3	0.1063	9/65
Error404-PG-4	0.0383	4/65
Error404-PO-1	0.0179	3/65
all other runs	0.0000	0/65

Appendix C. Per-stage cache statistics

Stage	Sonnet calls cached	Approx. cost (USD)
Stage 0 oracle (one-time)	65	\$2
Stage 1 candidate $\times$ 5 pools $\times$ 2 cal	650	\$15
Stage 1.5 Sonnet pointwise $\times$ 5 pools	12k	\$15
Stage 2 nuggets $\times$ 16 configurations	1k	\$5
Stage 3 bogus filter $\times$ 16 configurations	4k	\$10
Stage 4 verifier $\times$ 16 configurations	1k	\$5
Self-eval (Stage A + B) $\times$ 16 + ensemble	4k	\$15
RQ7 abstention sweep	0	\$0 (post-hoc)
RQ9 ensemble clustering	100	\$1
RQ10 Haiku ablation	1k	\$5 (mostly Haiku)
Total		~\$73